

Week 5

Keel Bone Chicken Project

06/15/26



College of Engineering

This Week

Weekly Progress

- Database Updates
 - MongoDB Development Continues
 - IDE Setup
- PCB Updates
 - Hardware Changes
- Sensor Testing Updates
 - More information about testing equipment

Database Updates

MongoDB Development Work Continues

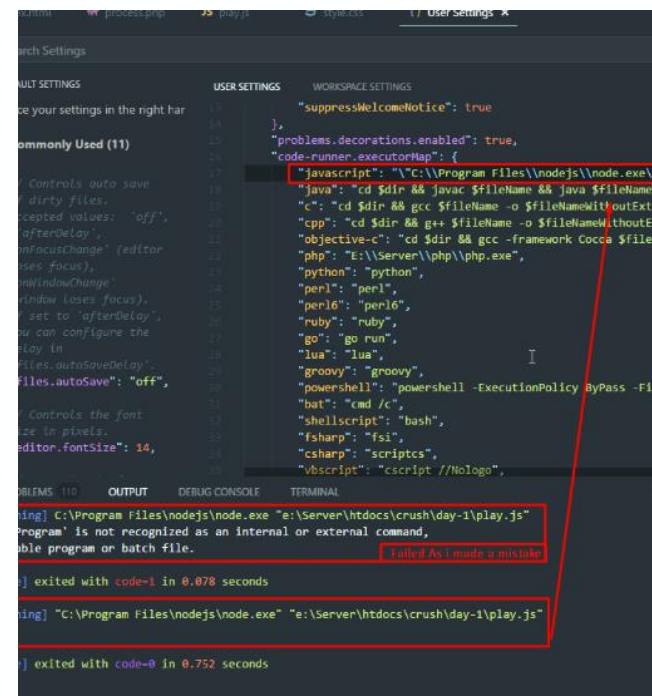
- Verified data from collection:
 - Seems to be fine but a slight issue with sampling
- Cluster Data and usage is acceptable
 - Graph below shows data capacity



Database Updates

IDE Setup

- IDE and code setup now properly mapped to MongoDB backend
- .ENV files match up with the results from the code
- Nicla Payloads should go directly to the backend



The screenshot shows the VS Code interface with the 'User Settings' tab active. The 'USER SETTINGS' section is expanded, showing a list of settings. The 'code-runner.executorMap' is configured with various programming languages and their corresponding executables. A red box highlights the 'javascript' entry, which is set to 'C:\Program Files\nodejs\node.exe'. Below the settings, the 'OUTPUT' window shows the execution of a command: 'C:\Program Files\nodejs\node.exe "e:\Server\htdocs\crush\day-1\play.js"'. The output indicates that the command failed with code 1, and a red box highlights the error message: 'Failed As I made a mistake'.

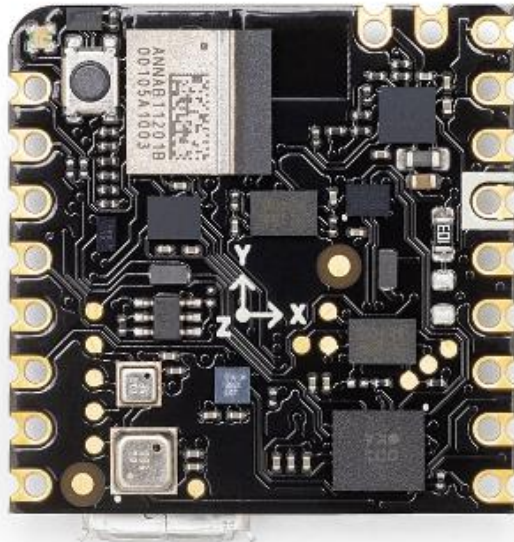
```
USER SETTINGS
"suppressWelcomeNotice": true
},
"problems.decorations.enabled": true,
"code-runner.executorMap": {
  "javascript": "C:\\Program Files\\nodejs\\node.exe",
  "java": "cd $dir && javac $fileName && java $fileName",
  "c": "cd $dir && gcc $fileName -o $fileNameWithoutExt",
  "cpp": "cd $dir && g++ $fileName -o $fileNameWithoutExt",
  "objective-c": "cd $dir && gcc -framework Cocoa $file",
  "php": "E:\\Server\\php\\php.exe",
  "python": "python",
  "perl": "perl",
  "perl6": "perl6",
  "ruby": "ruby",
  "go": "go run",
  "lua": "lua",
  "groovy": "groovy",
  "powershell": "powershell -ExecutionPolicy ByPass -File",
  "bat": "cmd /c",
  "shellscript": "bash",
  "fsharp": "fsharp",
  "csharp": "csc",
  "vbscript": "cscript //NoLogo",
  "batch": "cmd /c"
}
```

```
OUTPUT
[ting] C:\Program Files\nodejs\node.exe "e:\Server\htdocs\crush\day-1\play.js"
'program' is not recognized as an internal or external command,
operable program or batch file.
] exited with code=1 in 0.078 seconds
[ting] "C:\Program Files\nodejs\node.exe" "e:\Server\htdocs\crush\day-1\play.js"
] exited with code=0 in 0.752 seconds
```

PCB Updates

Hardware Changes

- Initial design rejected because of improper Gerber files
- New hardware implemented before final design



Nicla Sense ME Payload

The beta version currently transmits the following data structure to the local server:

```
{  
  temp: null, hum: null,  
  press: null, gas: null,  
  acc: {x,y,z}, gyro: {x,y,z}  
}
```

Constant BACKEND_URL set to localhost:5000 for stability during lab validation.

Sensor Testing Updates

More information about testing equipment

Current Lab Equipment:

- 50% - Basic Force

Prof. Wu's Equipment

- 95% - High Precision

Current lab force testing is insufficient for precise in-out calibration. Prof. Wenzhuo Wu's lab provides the specialized equipment needed for high-resolution force sensor validation.

- Contacted professor Wu, still waiting on a response

Next Week

Plan for next week

- Make gerber file and drill file revisions to PCBs for re-ordering
- Continue work on Database Development
 - ~3-4 more weeks to fully refine database
 - Look into hybrid option with infrequent backups (Would use same database model)
 - See if that option is feasible or a waste of resources
- Meet with Professor Wu about sensor testing

Thank You

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